

In class

1. Loading data

In [250... `!pip install nltk`

Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
 Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.2.1)
 Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.1)
 Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2024.11.6)
 Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)

In [251... `from google.colab import drive`
`drive.mount('/content/drive')`

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

In [252... `import os`
`file_path = '/content/drive/MyDrive/CPE393/books.txt'`
`print(os.path.exists(file_path))`

True

In [253... `import nltk`
`# nltk.download('all')`
`# nltk.download() #for the missing library`
`nltk.download('punkt_tab')`
`dir = '/content/drive/MyDrive/CPE393/'`
`f = open(dir+'books.txt', encoding='utf-8')`
`raw = f.read()`
`tokens = nltk.word_tokenize(raw)`

[nltk_data] Downloading package punkt_tab to /root/nltk_data...
 [nltk_data] Package punkt_tab is already up-to-date!

In [254... `print(tokens[1000:1100])`

['', 'eating', 'banquets', 'in', 'the', 'Great', 'Hall', ',', 'sleep\xading', 'i', 'n', 'his', 'four-poster', 'bed', 'in', 'the', 'tower', 'dormitory', ',', 'visitin', 'g', 'the', 'gamekeeper', ',', 'Hagrid', ',', 'in', 'his', 'cabin', 'next', 'to', 'the', 'Forbidden', 'Forest', 'in', 'the', 'grounds', ',', 'and', ',', 'especiall', 'y', ',', 'Quidditch', ',', 'the', 'most', 'popular', 'sport', 'in', 'the', 'wizar', 'ding', 'world', '(', 'six', 'tall', 'goal', 'posts', ',', 'four', 'flying', 'ball', 's', ',', 'and', 'four\xadteen', 'players', 'on', 'broomsticks', ')', '.', 'All', 'Harry', ',', 's', 'spellbooks', ',', 'his', 'wand', ',', 'robes', ',', 'cauldro', 'n', ',', 'and', 'top-of-the-line', 'Nimbus', 'Two', 'Thousand', 'broomstick', 'ha', 'd', 'been', 'locked', 'in', 'a', 'cupboard', 'under', 'the', 'stairs', 'by', 'Unc', 'le', 'Vernon', 'the', 'instant']

2. Regular expression

In [255... `tokens = [w.lower() for w in tokens]`

In [256... `import re`
`[w for w in tokens if re.search('ed$', w)][:10]`

Out[256... `['reserved',
 'published',
 'registered',
 'related',
 'reproduced',
 'stored',
 'transmitted',
 'opened',
 'released',
 'printed']`

1. What kind of input that matches, and what do not? Also, give a brief explanation.

Ans: `ed$` หมายถึง ค้นหาคำที่ลงท้ายด้วยตัวอักษร "ed"

ดังนั้นคำที่แสดงออกมาจะเป็นคำที่ลงท้ายด้วย "ed" เช่น reserved, published
 ในทางกลับกันคำที่ไม่ตรงกับรูปแบบนี้ก็คือคำที่ไม่ได้ลงท้ายด้วย "ed" เช่น

- education (มี ed อยู่ข้างหน้า ไม่ใช่ท้ายคำ)
- fast (ไม่ลงท้ายด้วย ed)

In [257... `[w for w in tokens if re.search('^.j..t..$', w)]`

Out[257... `['dejected',
 'adjusted',
 'rejected',
 'rejected',
 'dejected',
 'adjusted',
 'adjusted',
 'adjusted',
 'adjusted']`

2. What kind of input that matches, and what do not? Also, give a brief explanation.

Ans: `^.j..t..$` หมายถึง ค้นหาคำที่มี 8 ตัว ซึ่งตัวที่ 3 เป็น "j" และตัวที่ 6 เป็น "t" ส่วนตัวที่เลือกเป็นอะไรก็ได้

ดังนั้นคำที่แสดงออกมาจะเป็นคำที่มี 8 ตัว ซึ่งตัวที่ 3 เป็น "j" และตัวที่ 6 เป็น "t" ส่วนตัวที่เลือกเป็นอะไรก็ได้เช่น dejected, adjusted

ในทางกลับกันคำที่ไม่ตรงกับรูปแบบนี้ก็คือคำที่มีตัวอักษรไม่เท่ากับ 8 ตัวหรือตัวที่ 3 ไม่เป็น "j" และตัวที่ 6 ไม่เป็น "t"

- reject (ความยาวแค่ 6 ตัว)
- projected (ความยาวเกิน 8 ตัว)
- objected (ตำแหน่งที่ 3 ไม่ใช่ j)

In [258... `[w for w in tokens if re.search('^spo.ts?$', w)][:10]`

Out[258... `['sport',
'sports',
'sport',
'sport',
'sports',
'sport',
'sports',
'sports',
'sports',
'sports']`

3. What kind of input that matches, and what do not? Also, give a brief explanation.

Ans: `^spo.ts?$` หมายถึง ค้นหาคำที่ขึ้นต้นด้วย "spo" ตามด้วยตัวอักษรใด ๆ หนึ่งตัว และตามด้วย "t" จากนั้นอาจจะมี "s" 1 ตัวหรือไม่มีเลยก็ได้

ดังนั้นคำที่ตรงกับรูปแบบนี้จะเป็นคำที่มีความยาว 5 หรือ 6 ตัวอักษร เช่น sport, spots

ในทางกลับกัน คำที่ไม่ตรงคือคำที่ไม่ได้ขึ้นต้นด้วย "spo" หรือไม่มี "t" เป็นตัวรองสุดท้าย เช่น

- spoon (ไม่มีตัว t)
- spore (ไม่มีตัว t)
- sporting (เกิน 6 ตัวอักษร)

In [259... `[w for w in tokens if re.search('^[b-f]', w)][:10]`

Out[259... `['chamber',
'by',
'by',
'books',
'for',
'f.',
'driver',
'foul-weather',
'friend',
'copyright']`

4. What kind of input that matches, and what do not? Also, give a brief explanation.

Ans: `^[b-f]` หมายถึง ค้นหาคำที่ขึ้นต้นด้วยตัวอักษรใดก็ได้ระหว่าง b ถึง f (รวม b, c, d, e, f)

ดังนั้นคำที่ตรงกับรูปแบบนี้คือคำที่อักษรตัวแรกอยู่ในช่วง b–f เช่น banana, cat, dog, elephant, fish

ในทางกลับกันคำที่ไม่ตรงคือคำที่ไม่ได้ขึ้นต้นด้วย b–f เช่น

- apple (ขึ้นต้นด้วย a)
- grape (ขึ้นต้นด้วย g)
- zebra (ขึ้นต้นด้วย z)

In [260...

```
[w for w in tokens if re.search('^[bdf][aue][rts]$', w)][:10]
```

Out[260...

```
['but', 'but', 'far', 'but', 'but', 'but', 'fat', 'but', 'fat', 'but']
```

5. What kind of input that matches, and what do not? Also, give a brief explanation.

Ans: `^[bdf][aue][rts]$` หมายถึง ค้นหาคำที่มีความยาว 3 ตัวอักษร โดยตัวแรกต้องเป็น **b, d หรือ f** ตัวที่สองต้องเป็น **a, u หรือ e** และตัวที่สามต้องเป็น **r, t หรือ s**

ดังนั้นคำที่ตรงกับรูปแบบนี้คือคำที่มี 3 ตัวอักษร และตรงตามเงื่อนไขแต่ละตัว เช่น but, bet, fur, dus

ในทางกลับกันคำที่ไม่ตรงคือคำที่ไม่ใช่ 3 ตัวอักษร หรือไม่ตรงตามเงื่อนไขตัวอักษรแต่ละตำแหน่ง เช่น

- cat (ตัวแรกไม่ใช่ b, d, f)
- fun (ตัวสามไม่ใช่ r, t, s)
- dusted (ความยาวเกิน 3 ตัว)

In [261...

```
set([w for w in tokens if re.search('noo+$', w)])
```

Out[261...

```
{'nooo',
 'noooo',
 'nooooo',
 'noooooo',
 'nooooooo',
 'noooooooo',
 'nooooooooo',
 'nooooooooooo',
 'noooooooooooo',
 'nooooooooooooo'}
```

6. What kind of input that matches, and what do not? Also, give a brief explanation.

Ans: `noo+$` หมายถึง ค้นหาคำที่ลงท้ายด้วยตัวอักษร "no" ตามด้วย "o" ซ้ำอย่างน้อยหนึ่งตัว

ดังนั้นคำที่ตรงกับรูปแบบนี้คือคำที่ลงท้ายด้วย "no" + "o" ซ้ำ เช่น nooo, nooooo, technoo

ในทางกลับกันคำที่ไม่ตรงคือคำที่ไม่ลงท้ายด้วย "no" + "o" ซ้ำ เช่น

- no (ไม่ลงท้ายด้วย "no" + "o" อย่างน้อยหนึ่งตัว)
- noon (ไม่ลงท้ายด้วย "no" + "o" อย่างน้อยหนึ่งตัว)

- banana (ไม่ลงท้ายด้วย "no" + "o" อย่างน้อยหนึ่งตัว)

```
In [262... set([w for w in tokens if re.search('noo*$', w)])
```

```
Out[262... {'albino',
              'dunno',
              'gemino',
              'inferno',
              'n-no',
              'no',
              'nooo',
              'noooo',
              'nooooo',
              'noooooo',
              'nooooooo',
              'noooooooo',
              'nooooooooo',
              'nooooooooooooo',
              'nooooooooooooooooo',
              'oppugno',
              'palomino',
              'patrono',
              'piano',
              'pi\xadano',
              'rhino',
              'xeno'}
```

7. What kind of input that matches, and what do not? Also, give a brief explanation.

Ans: `noo*$` หมายถึง ค้นหาคำที่ลงท้ายด้วยตัวอักษร "no" แล้วตามด้วย "o" ตั้งแต่ 0 ตัวขึ้นไป

ดังนั้นค่าที่ตรงกับรูปแบบนี้คือค่าที่ลงท้ายด้วย "no" แล้วตามด้วย "o" ตั้งแต่ 0 ตัวขึ้นไป เช่น no, noo, nooo, techno

ในทางกลับกันค่าที่ไม่ตรงคือค่าที่ไม่ลงท้ายด้วย "no" แล้วตามด้วย "o" ตั้งแต่ 0 ตัวขึ้นไป เช่น

- na (ไม่ลงท้ายด้วย "no" แล้วตามด้วย "o" ตั้งแต่ 0 ตัวขึ้นไป)
- banana (ไม่ลงท้ายด้วย "no" แล้วตามด้วย "o" ตั้งแต่ 0 ตัวขึ้นไป)
- note (ไม่ลงท้ายด้วย "no" แล้วตามด้วย "o" ตั้งแต่ 0 ตัวขึ้นไป)

```
In [263... [w for w in tokens if re.search('\.+$', w)][:10]
```

```
<>:1: SyntaxWarning: invalid escape sequence '\.'  
<>:1: SyntaxWarning: invalid escape sequence '\.'  
/tmp/ipython-input-707925584.py:1: SyntaxWarning: invalid escape sequence '\.'  
[w for w in tokens if re.search('\.+$', w)][10]
```

```
Out[263]: ['j.', 'k.', 'a.', 'p.', 'f.', '.', 'j.', 'k.', '.', 'bros.']
```

8. What kind of input that matches, and what do not? Also, give a brief explanation.

Ans: `.$` หมายถึง ค้นหาคำหรือสตริงที่ลงท้ายด้วยจุด (.) อย่างน้อย 1 ตัว
 ดังนั้นคำที่ตรงกับรูปแบบนี้คือคำหรือสตริงที่ลงท้ายด้วย "." เช่น ., end., hello..., version.
 ในทางกลับกันคำที่ไม่ตรงคือคำหรือสตริงที่ไม่ลงท้ายด้วย "." เช่น

- .hello (ไม่มีจุดลงท้าย)
- world (ไม่มีจุดลงท้าย)
- test (ไม่มีจุดลงท้าย)

In [264...

```
[w for w in tokens if re.search('^[A-Z]+\.$', w)]
```

```
<>:1: SyntaxWarning: invalid escape sequence '\.'
<>:1: SyntaxWarning: invalid escape sequence '\.'
/tmp/ipython-input-1844296225.py:1: SyntaxWarning: invalid escape sequence '\.'
[w for w in tokens if re.search('^[A-Z]+\.$', w)]
```

Out[264...

```
[]
```

9. What kind of input that matches, and what do not? Also, give a brief explanation.

Ans: `^[A-Z]+.` หมายถึง ค้นหาคำที่ประกอบด้วยตัวอักษรพิมพ์ใหญ่ (A-Z) อย่างน้อยหนึ่งตัว และลงท้ายด้วยจุด (.)

ดังนั้นคำที่ตรงกับรูปแบบนี้คือคำที่เป็นตัวพิมพ์ใหญ่ทั้งหมดและลงท้ายด้วย "." เช่น HELLO., WORLD., TEST.

ในทางกลับกันคำที่ไม่ตรงคือคำหรือสตริงที่ไม่ใช่ตัวพิมพ์ใหญ่ทั้งหมด หรือไม่ลงท้ายด้วย "." เช่น

- Hello. (มีตัวพิมพ์เล็ก)
- WORLD (ไม่มีจุดลงท้าย)
- test. (เป็นตัวพิมพ์เล็ก)

In [265...

```
[w for w in tokens if re.search('^[0-9]{4}$', w)][1:10]
```

Out[265...

```
['1999',
 '1999',
 '1920',
 '1999',
 '1999',
 '1875',
 '1492',
 '1289',
 '2007',
 '2007']
```

10. What kind of input that matches, and what do not? Also, give a brief explanation.

Ans: `^[0-9]{4}$` หมายถึง ค้นหาคำที่ประกอบด้วยตัวเลข 0-9 จำนวน 4 ตัว

ดังนั้นคำที่ตรงกับรูปแบบนี้คือตัวเลข 4 ตัว เช่น 2025, 1234, 0001

ในทางกลับกันคำที่ไม่ตรงคือสตริงที่ไม่ใช่ตัวเลข 4 ตัว เช่น

- 123 (มีแค่ 3 ตัว)
- 12345 (มี 5 ตัว)
- ab12 (มีตัวอักษร)

In [266... `set([w for w in tokens if re.search('(ed|ing)$', w)][0:10])`

Out[266... `{'published',
'publishing',
'registered',
'related',
'reproduced',
'reserved',
'rowling',
'stored'}`

11. What kind of input that matches, and what do not? Also, give a brief explanation.

Ans: **(ed|ing)\$** หมายถึง ค้นหาคำที่ลงท้ายด้วย "ed" หรือ "ing"

ดังนั้นคำที่ตรงกับรูปแบบนี้คือคำที่ลงท้ายด้วย "ed" หรือ "ing" เช่น walked, jumping, played, running

ในทางกลับกันคำที่ไม่ตรงคือคำที่ไม่ลงท้ายด้วย "ed" หรือ "ing" เช่น

- walk (ไม่มี "ed" หรือ "ing")
- run (ไม่มี "ed" หรือ "ing")
- play (ไม่มี "ed" หรือ "ing")

3. Regular expression tokenizer

3.1 Basic regex tokenizer

In [267... `s = ("Good muffins cost $3.88\nin New York. Please buy me\n"
"two of them.\n\nThanks.")
s2 = ("Alas, it has not rained today. When, do you think, "
"will it rain again?")
s3 = ("<p>Although this is not the case here, we must "
"not relax our vigilance!</p>")`

In [268... `s2`

Out[268... `'Alas, it has not rained today. When, do you think, will it rain again?'`

In [269... `nltk.regexp_tokenize(s2, r'[,.\?!"]\s*', gaps=True)`

Out[269... `['Alas',
'it has not rained today',
'When',
'do you think',
'will it rain again']`

12.What happened after applying the above RegEx tokenizer to the input string?

Ans: แยกคำโดยใช้เครื่องหมาย , . ? ! " เป็นตัวแบ่งคำ

In [270...

s3

Out[270...

```
'<p>Although this is <b>not</b> the case here, we must not relax our vigilance!
</p>'
```

In [271...

```
nltk.regex_tokenize(s3, r'</?.>', gaps=False)
```

Out[271...

```
['<p>', '<b>', '</b>', '</p>']
```

13.What happened after applying the above RegEx tokenizer to the input string?

Ans: ค้นหาคำที่มีเครื่องหมาย < > ประกบ เช่น ,

In [272...

```
nltk.regex_tokenize(s3, r'</?.>', gaps=True)
```

Out[272...

```
['Although this is ',
 'not',
 ' the case here, we must not relax our vigilance!']
```

14.What happened after applying the above RegEx tokenizer to the input string?

Ans: **แยกคำโดยใช้เครื่องหมาย < > เป็นตัวแบ่งคำ****

In [273...

```
tokens[:10]
```

Out[273...

```
['harry', 'potter', 'and', 'the', 'chamber', 'of', 'secrets', 'by', 'j.', 'k.']
```

Harry Potter series

Cleaning text

In [274...

```
# 1. Lowercase all tokens
tokens = [w.lower() for w in tokens]
```

In [275...

```
# 2. Show punctuation and non-alphanumeric characters
from collections import Counter
punctuation_chars = []

# find
for w in tokens:
    punctuation_chars.extend(re.findall(r'^a-zA-Z0-9]', w))
```



```
# counter
punctuation_count = Counter(punctuation_chars)

# result
for char, count in punctuation_count.items():
    print(f"{char} : {count}")
```

```
. : 65759
é : 30
á : 1
- : 5660
@ : 23
, : 86310
/ : 10
: : 1474
[ : 6
- : 9711
] : 6
' : 34979
“ : 37060
! : 7637
” : 36750
? : 10865
: : 9622
... : 7957
; : 3482
‘ : 511
( : 329
) : 331
* : 57
' : 3
& : 2
” : 1
ß : 1
¬ : 1
- : 9
ù : 1
```

```
In [276... # 3. Remove punctuation and non-alphanumeric characters from each token
tokens = [re.sub(r'^a-zA-Z0-9', '', w) for w in tokens]
```

```
In [289... # 4. Show space or empty tokens
empty_count = sum(w.strip() == '' for w in tokens)
print(f"Empty or space-only tokens count: {empty_count}")
```

Empty or space-only tokens count: 39878

```
In [290... # 5. Remove space or empty tokens
tokens = [w for w in tokens if re.sub(r'\s+', '', w, flags=re.UNICODE) != '']
```

```
In [279... # 6. Show single characters b-z
single_chars = []

# find
for w in tokens:
    single_chars.extend(re.findall(r'^[a-z]{1}$', w))

# counter
single_count = Counter(single_chars)
```

```
# result
for char, count in sorted(single_count.items()):
    print(f"'{char}' : {count}")
```

```
'a' : 21038
'b' : 13
'c' : 61
'd' : 1984
'e' : 52
'f' : 5
'g' : 5
'h' : 9
'i' : 13557
'j' : 22
'k' : 19
'l' : 4
'm' : 1358
'n' : 25
'o' : 258
'p' : 15
'r' : 7
's' : 14750
't' : 7703
'u' : 1
'v' : 1
'w' : 6
'x' : 2
'y' : 20
'z' : 1
```

```
In [280... # 7. Remove single characters b-z
tokens = [re.sub(r'\b[b-z]\b', '', w) for w in tokens]
```

```
In [291... print("Clean tokens:", tokens[:10])
```

Clean tokens: ['harry', 'potter', 'and', 'the', 'chamber', 'of', 'secrets', 'by', 'rowling', 'illustrations']

EDA

จำนวนบทของหนังสือ

```
In [292... cleaned_text = ' '.join(tokens) # รวม tokens เป็น string

# ใช้ regex chapter [\d][\d]* → chapter ตามด้วยตัวเลข 1 หลักขึ้นไป
chapter_matches = re.findall(r'chapter [\d][\d]*', cleaned_text)

# นับจำนวน occurrence
chapter_count = len(chapter_matches)

print(f"The phrase 'chapter xx' appears {chapter_count} times in the text.")
```

The phrase 'chapter xx' appears 199 times in the text.

จำนวนเล่มของหนังสือ

```
In [293... # ใช้ regex หา "chapter 1"
chapter_1_matches = re.findall(r'\bchapter 1\b', cleaned_text)
```

```
# นับจำนวน occurrence
chapter_1_count = len(chapter_1_matches)

print(f"The phrase 'chapter 1' appears {chapter_1_count} times in the text.")
```

The phrase 'chapter 1' appears 7 times in the text.

In [294...

```
def extract_chapters(book):
    chapters_in_book = re.findall(r'chapter \d+\b', book, flags=re.IGNORECASE)

    start = 0
    while(start != len(chapters_in_book)):
        chap_id = "chapter {}".format(start+1)
        start_index = cleaned_text.index(chap_id)
        print(cleaned_text[start_index:start_index+100])

        start += 1

book_sections = re.split(r'(?i)(?=chapter \d\b)', cleaned_text)
books = [section.strip() for section in book_sections if section.strip() != '']

for i, book in enumerate(books[1:]): # books[0] is just preface so, skipped!
    print(f'=====')
    print(f'Book {i+1} content')
    extract_chapters(book)
    print("\n")
```

=====

Book 1 content

chapter 1 the worst birthday not for the first time an argument had broken out over breakfast at num
chapter 2 dobby warning harry managed not to shout out but it was a close thing the little creature
chapter 3 the burrow ron breathed harry creeping to the window and pushing it up so they could talk
chapter 4 at flourish and blotts life at the burrow was as different as possible from life on privet
chapter 5 the whomping willow the end of the summer vacation came too quickly for harry liking he wa
chapter 6 gilderoy lockhart the next day however harry barely grinned once things started to go down
chapter 7 mudbloods and murmurs harry spent a lot of time over the next few days dodging out of sigh
chapter 8 the deathday party october arrived spreading a damp chill over the grounds and into the ca
chapter 9 the writing on the wall what going on here what going on attracted no doubt by malfoy shou
chapter 10 the rogue bludger since the disastrous episode of the pixies professor lockhart had not b
chapter 11 the dueling club harry woke up on sunday morning to find the dormitory blazing with winte
chapter 12 the polyjuice potion they stepped off the stone staircase at the top and professor mcgona
chapter 13 the very secret diary hermione remained in the hospital wing for several weeks there was
chapter 14 cornelius fudge harry ron and hermione had always known that hagrid had an unfortunate li
chapter 15 aragog summer was creeping over the grounds around the castle sky and lake alike turned p
chapter 16 the chamber of secrets all those times we were in that bathroom and she was just three to
chapter 17 the heir of slytherin he was standing at the end of a very long dimly lit chamber towerin
chapter 18 dobby reward for a moment there was silence as harry ron ginny and lockhart stood in the

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Book 2 content

chapter 1 the worst birthday not for the first time an argument had broken out over breakfast at num
chapter 2 dobby warning harry managed not to shout out but it was a close thing the little creature
chapter 3 the burrow ron breathed harry creeping to the window and pushing it up so they could talk
chapter 4 at flourish and blotts life at the burrow was as different as possible from life on privet
chapter 5 the whomping willow the end of the summer vacation came too quickly for harry liking he wa
chapter 6 gilderoy lockhart the next day however harry barely grinned once things started to go down
chapter 7 mudbloods and murmurs harry spent a lot of time over the next few days dodging out of sigh
chapter 8 the deathday party october arrived spreading a damp chill over the grounds and into the ca
chapter 9 the writing on the wall what going on here what going on attracted no doubt by malfoy shou

chapter 10 the rogue bludger since the disastrous episode of the pixies professor lockhart had not b
chapter 11 the dueling club harry woke up on sunday morning to find the dormitory blazing with winte
chapter 12 the polyjuice potion they stepped off the stone staircase at the top a nd professor mcgona
chapter 13 the very secret diary hermione remained in the hospital wing for sever al weeks there was
chapter 14 cornelius fudge harry ron and hermione had always known that hagrid ha d an unfortunate li
chapter 15 aragog summer was creeping over the grounds around the castle sky and lake alike turned p
chapter 16 the chamber of secrets all those times we were in that bathroom and sh e was just three to
chapter 17 the heir of slytherin he was standing at the end of a very long dimly lit chamber towerin
chapter 18 dobby reward for a moment there was silence as harry ron ginny and loc khart stood in the
chapter 19 the silver doe it was snowing by the time hermione took over the watch at midnight harry
chapter 20 xenophilius lovegood harry had not expected hermione anger to abate ov ernight and was the
chapter 21 the tale of the three brothers harry turned to look at ron and hermion e neither of them s
chapter 22 what caused this abrupt rupture had dumbledore come to his senses had he told grindelwald
chapter 23 malfoy manor harry looked around at the other two now mere outlines in the darkness he sa
chapter 24 the wandmaker it was like sinking into an old nightmare for an instant harry knelt again
chapter 25 shell cottage bill and fleur cottage stood alone on a cliff overlookin g the sea its walls
chapter 26 gringotts their plans were made their preparations complete in the sma llest bedroom a sin
chapter 27 the final hiding place there was no means of steering the dragon could not see where it w
chapter 28 the missing mirror harry feet touched road he saw the achingly familia r hogsmeade high st
chapter 29 the lost diadem neville what the how but neville had spotted ron and h ermione and with ye
chapter 30 the sacking of severus snape the moment her finger touched the mark ha rry scar burned sav
chapter 31 the battle of hogwarts the enchanted ceiling of the great hall was dar k and scattered wit
chapter 32 the elder wand the world had ended so why had the battle not ceased th e castle fallen sil
chapter 33 the prince tale harry remained kneeling at snape side simply staring d own at him until qu
chapter 34 the forest again finally the truth lying with his face pressed into th e dusty carpet of t
chapter 35 king cross he lay facedown listening to the silence he was perfectly a lone nobody was wat
chapter 36 the flaw in the plan he was lying facedown on the ground again the sme ll of the forest fi
chapter 37 the beginning when he looked back even a month later harry found he ha d only scattered me

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Book 3 content

chapter 1 the worst birthday not for the first time an argument had broken out over breakfast at num

chapter 2 dobby warning harry managed not to shout out but it was a close thing the little creature

chapter 3 the burrow ron breathed harry creeping to the window and pushing it up so they could talk

chapter 4 at flourish and blotts life at the burrow was as different as possible from life on privet

chapter 5 the whomping willow the end of the summer vacation came too quickly for harry liking he wa

chapter 6 gilderoy lockhart the next day however harry barely grinned once things started to go down

chapter 7 mudbloods and murmurs harry spent a lot of time over the next few days dodging out of sigh

chapter 8 the deathday party october arrived spreading a damp chill over the grounds and into the ca

chapter 9 the writing on the wall what going on here what going on attracted no doubt by malfoy shou

chapter 10 the rogue bludger since the disastrous episode of the pixies professor lockhart had not b

chapter 11 the dueling club harry woke up on sunday morning to find the dormitory blazing with winte

chapter 12 the polyjuice potion they stepped off the stone staircase at the top and professor mcgona

chapter 13 the very secret diary hermione remained in the hospital wing for several weeks there was

chapter 14 cornelius fudge harry ron and hermione had always known that hagrid had an unfortunate li

chapter 15 aragog summer was creeping over the grounds around the castle sky and lake alike turned p

chapter 16 the chamber of secrets all those times we were in that bathroom and she was just three to

chapter 17 the heir of slytherin he was standing at the end of a very long dimly lit chamber towerin

chapter 18 dobby reward for a moment there was silence as harry ron ginny and lockhart stood in the

chapter 19 the silver doe it was snowing by the time hermione took over the watch at midnight harry

chapter 20 xenophilius lovegood harry had not expected hermione anger to abate overnight and was the

chapter 21 the tale of the three brothers harry turned to look at ron and hermione neither of them s

chapter 22 what caused this abrupt rupture had dumbledore come to his senses had he told grindelwald

chapter 23 malfoy manor harry looked around at the other two now mere outlines in the darkness he sa

chapter 24 the wandmaker it was like sinking into an old nightmare for an instant harry knelt again

chapter 25 shell cottage bill and fleur cottage stood alone on a cliff overlooking the sea its walls

chapter 26 gringotts their plans were made their preparations complete in the smallest bedroom a sin

chapter 27 the final hiding place there was no means of steering the dragon could not see where it w

chapter 28 the missing mirror harry feet touched road he saw the achingly familiar hogsmeade high st

chapter 29 the lost diadem neville what the how but neville had spotted ron and hermione and with ye

chapter 30 the sacking of severus Snape the moment her finger touched the mark harry scar burned sav

chapter 31 the battle of hogwarts the enchanted ceiling of the great hall was dark and scattered with
 chapter 32 the elder wand the world had ended so why had the battle not ceased the castle fallen silent
 chapter 33 the prince tale harry remained kneeling at snake side simply staring down at him until
 chapter 34 the forest again finally the truth lying with his face pressed into the dusty carpet of the
 chapter 35 king's cross he lay facedown listening to the silence he was perfectly alone nobody was watching
 chapter 36 the flaw in the plan he was lying facedown on the ground again the smell of the forest fire
 chapter 37 the beginning when he looked back even a month later harry found he had only scattered memories

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Book 4 content

chapter 1 the worst birthday not for the first time an argument had broken out over breakfast at number
 chapter 2 dobby warning harry managed not to shout out but it was a close thing to the little creature
 chapter 3 the burrow ron breathed harry creeping to the window and pushing it up so they could talk
 chapter 4 at flourish and blott's life at the burrow was as different as possible from life on Privet
 chapter 5 the whomping willow the end of the summer vacation came too quickly for harry liking he was
 chapter 6 gilderoy lockhart the next day however harry barely grinned once things started to go down
 chapter 7 mudbloods and murmurs harry spent a lot of time over the next few days dodging out of sight
 chapter 8 the deathday party october arrived spreading a damp chill over the grounds and into the castle
 chapter 9 the writing on the wall what going on here what going on attracted no doubt by malfoy's shout
 chapter 10 the rogue bludger since the disastrous episode of the pixies professor lockhart had not been
 chapter 11 the dueling club harry woke up on sunday morning to find the dormitory blazing with winter
 chapter 12 the polyjuice potion they stepped off the stone staircase at the top and professor mcgonagall
 chapter 13 the very secret diary hermione remained in the hospital wing for several weeks there was
 chapter 14 cornelius fudge harry ron and hermione had always known that hagrid had an unfortunate life
 chapter 15 aragog summer was creeping over the grounds around the castle sky and lake alike turned purple
 chapter 16 the chamber of secrets all those times we were in that bathroom and she was just three to
 chapter 17 the heir of slytherin he was standing at the end of a very long dimly lit chamber tower in
 chapter 18 dobby reward for a moment there was silence as harry ron ginny and lockhart stood in the
 chapter 19 the silver doe it was snowing by the time hermione took over the watch at midnight harry
 chapter 20 xenophilius lovegood harry had not expected hermione's anger to abate overnight and was the
 chapter 21 the tale of the three brothers harry turned to look at ron and hermione neither of them said

chapter 22 what caused this abrupt rupture had dumbledore come to his senses had he told grindelwald
 chapter 23 malfoy manor harry looked around at the other two now mere outlines in the darkness he sa
 chapter 24 the wandmaker it was like sinking into an old nightmare for an instant harry knelt again
 chapter 25 shell cottage bill and fleur cottage stood alone on a cliff overlooking the sea its walls
 chapter 26 gringotts their plans were made their preparations complete in the smallest bedroom a sin
 chapter 27 the final hiding place there was no means of steering the dragon could not see where it w
 chapter 28 the missing mirror harry feet touched road he saw the achingly familiar hogsmeade high st
 chapter 29 the lost diadem neville what the how but neville had spotted ron and hermione and with ye
 chapter 30 the sacking of severus snape the moment her finger touched the mark harry scar burned sav

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Book 5 content

chapter 1 the worst birthday not for the first time an argument had broken out over breakfast at num
 chapter 2 dobby warning harry managed not to shout out but it was a close thing the little creature
 chapter 3 the burrow ron breathed harry creeping to the window and pushing it up so they could talk
 chapter 4 at flourish and blotts life at the burrow was as different as possible from life on privet
 chapter 5 the whomping willow the end of the summer vacation came too quickly for harry liking he wa
 chapter 6 gilderoy lockhart the next day however harry barely grinned once things started to go down
 chapter 7 mudbloods and murmurs harry spent a lot of time over the next few days dodging out of sigh
 chapter 8 the deathday party october arrived spreading a damp chill over the grounds and into the ca
 chapter 9 the writing on the wall what going on here what going on attracted no doubt by malfoy shou
 chapter 10 the rogue bludger since the disastrous episode of the pixies professor lockhart had not b
 chapter 11 the dueling club harry woke up on sunday morning to find the dormitory blazing with winte
 chapter 12 the polyjuice potion they stepped off the stone staircase at the top and professor mcgona
 chapter 13 the very secret diary hermione remained in the hospital wing for several weeks there was
 chapter 14 cornelius fudge harry ron and hermione had always known that hagrid had an unfortunate li
 chapter 15 aragog summer was creeping over the grounds around the castle sky and lake alike turned p
 chapter 16 the chamber of secrets all those times we were in that bathroom and she was just three to
 chapter 17 the heir of slytherin he was standing at the end of a very long dimly lit chamber towerin
 chapter 18 dobby reward for a moment there was silence as harry ron ginny and lockhart stood in the
 chapter 19 the silver doe it was snowing by the time hermione took over the watch at midnight harry

chapter 20 xenophilius lovegood harry had not expected hermione anger to abate overnight and was the
 chapter 21 the tale of the three brothers harry turned to look at ron and hermione neither of them
 chapter 22 what caused this abrupt rupture had dumbledore come to his senses had he told grindelwald
 chapter 23 malfoy manor harry looked around at the other two now mere outlines in the darkness he saw
 chapter 24 the wandmaker it was like sinking into an old nightmare for an instant harry knelt again
 chapter 25 shell cottage bill and fleur cottage stood alone on a cliff overlooking the sea its walls
 chapter 26 gringotts their plans were made their preparations complete in the smallest bedroom a simple
 chapter 27 the final hiding place there was no means of steering the dragon could not see where it was
 chapter 28 the missing mirror harry feet touched road he saw the achingly familiar hogsmeade high street
 chapter 29 the lost diadem neville what the how but neville had spotted ron and hermione and with them
 chapter 30 the sacking of severus snape the moment her finger touched the mark harry scar burned savagely
 chapter 31 the battle of hogwarts the enchanted ceiling of the great hall was dark and scattered with
 chapter 32 the elder wand the world had ended so why had the battle not ceased the castle fallen silent
 chapter 33 the prince tale harry remained kneeling at snape side simply staring down at him until quiet
 chapter 34 the forest again finally the truth lying with his face pressed into the dusty carpet of the forest
 chapter 35 king cross he lay facedown listening to the silence he was perfectly alone nobody was watching
 chapter 36 the flaw in the plan he was lying facedown on the ground again the smell of the forest fire
 chapter 37 the beginning when he looked back even a month later harry found he had only scattered memories
 chapter 38 the second war begins hewhomustnotbenamed returns in a brief statement friday night minis

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Book 6 content

chapter 1 the worst birthday not for the first time an argument had broken out over breakfast at number 12
 chapter 2 dobby warning harry managed not to shout out but it was a close thing the little creature
 chapter 3 the burrow ron breathed harry creeping to the window and pushing it up so they could talk
 chapter 4 at flourish and blotts life at the burrow was as different as possible from life on privet
 chapter 5 the whomping willow the end of the summer vacation came too quickly for harry liking he was
 chapter 6 gilderoy lockhart the next day however harry barely grinned once things started to go down
 chapter 7 mudbloods and murmurs harry spent a lot of time over the next few days dodging out of sight
 chapter 8 the deathday party october arrived spreading a damp chill over the grounds and into the castle
 chapter 9 the writing on the wall what going on here what going on attracted no doubt by malfoy shouting

chapter 10 the rogue bludger since the disastrous episode of the pixies professor lockhart had not b
 chapter 11 the dueling club harry woke up on sunday morning to find the dormitory blazing with winte
 chapter 12 the polyjuice potion they stepped off the stone staircase at the top a nd professor mcgona
 chapter 13 the very secret diary hermione remained in the hospital wing for sever al weeks there was
 chapter 14 cornelius fudge harry ron and hermione had always known that hagrid ha d an unfortunate li
 chapter 15 aragog summer was creeping over the grounds around the castle sky and lake alike turned p
 chapter 16 the chamber of secrets all those times we were in that bathroom and sh e was just three to
 chapter 17 the heir of slytherin he was standing at the end of a very long dimly lit chamber towerin
 chapter 18 dobby reward for a moment there was silence as harry ron ginny and loc khart stood in the
 chapter 19 the silver doe it was snowing by the time hermione took over the watch at midnight harry
 chapter 20 xenophilius lovegood harry had not expected hermione anger to abate ov ernight and was the
 chapter 21 the tale of the three brothers harry turned to look at ron and hermion e neither of them s
 chapter 22 what caused this abrupt rupture had dumbledore come to his senses had he told grindelwald

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Book 7 content

chapter 1 the worst birthday not for the first time an argument had broken out ov er breakfast at num
 chapter 2 dobby warning harry managed not to shout out but it was a close thing t he little creature
 chapter 3 the burrow ron breathed harry creeping to the window and pushing it up so they could talk
 chapter 4 at flourish and blotts life at the burrow was as different as possible from life on privet
 chapter 5 the whomping willow the end of the summer vacation came too quickly for harry liking he wa
 chapter 6 gilderoy lockhart the next day however harry barely grinned once things started to go down
 chapter 7 mudbloods and murmurs harry spent a lot of time over the next few days dodging out of sigh
 chapter 8 the deathday party october arrived spreading a damp chill over the grou nds and into the ca
 chapter 9 the writing on the wall what going on here what going on attracted no d oubt by malfoy shou
 chapter 10 the rogue bludger since the disastrous episode of the pixies professor lockhart had not b
 chapter 11 the dueling club harry woke up on sunday morning to find the dormitory blazing with winte
 chapter 12 the polyjuice potion they stepped off the stone staircase at the top a nd professor mcgona
 chapter 13 the very secret diary hermione remained in the hospital wing for sever al weeks there was
 chapter 14 cornelius fudge harry ron and hermione had always known that hagrid ha d an unfortunate li
 chapter 15 aragog summer was creeping over the grounds around the castle sky and lake alike turned p

chapter 16 the chamber of secrets all those times we were in that bathroom and she was just three to
chapter 17 the heir of slytherin he was standing at the end of a very long dimly lit chamber towerin

พิจารณา book_2 เนื่องจากจำนวนบทไม่ตรง เพราะ chapter 18 อ้างอิง chapter 22 ในเนื้อหา

หมายเหตุ!

ตอนแรกคิดว่าจะรวม chapter 18 กับ chapter 19 แล้วลบ chapter 19 ออกเหมือนใน comment

In [295...

```
book_index = 2 # enumerate start from 0
book_2 = books[book_index]

# split chapters
chapters_in_book2 = re.split(r'(?i)(?=chapter \d+\b)', book_2)
chapters_in_book2 = [ch.strip() for ch in chapters_in_book2 if ch.strip()]

# merge chapter 18 and 19, then drop chapter 19
# if len(chapters_in_book2) >= 19:
#     chapters_in_book2[17] = chapters_in_book2[17] + "\n\n" + chapters_in_book2
#     del chapters_in_book2[18] # remove chapter 19

# results
for j, chapter in enumerate(chapters_in_book2, 1):
    print(f"chapter {j}: {chapter[:100]}...") # preview first 100 char
```

chapter 1: chapter 1 the dark lord ascending the two men appeared out of nowhere a few yards apart in the narrow...

chapter 2: chapter 2 in memoriam harry was bleeding clutching his right hand in his left and swearing under his...

chapter 3: chapter 3 the dursleys departing the sound of the front door slamming echoed up the stairs and a voice...

chapter 4: chapter 4 the seven potters harry ran back upstairs to his bedroom arriving at the window just in time...

chapter 5: chapter 5 fallen warrior hagrid harry struggled to raise himself out of the debris of metal and leather...

chapter 6: chapter 6 the ghoul in pajamas the shock of losing mad-eye hung over the house in the days that followed...

chapter 7: chapter 7 the will of albus Dumbledore he was walking along a mountain road in the cool blue light of...

chapter 8: chapter 8 the wedding three o'clock on the following afternoon found harry, ron, fred and george standing...

chapter 9: chapter 9 a place to hide everything seemed fuzzy slow harry and hermi-one jumped to their feet and down...

chapter 10: chapter 10 Kreacher told harry woke early next morning wrapped in a sleeping bag on the drawing room...

chapter 11: chapter 11 the bribe if Kreacher could escape a lake full of inferi harry was confident that the cap...

chapter 12: chapter 12 magic is might as August wore on the square of unkempt grass in the middle of Grimmauld Place...

chapter 13: chapter 13 the Muggle-born Registration Commission ah, Malfoy said, looking at Hermione, travel...

chapter 14: chapter 14 the thief harry opened his eyes and was dazzled by gold and green he had no idea what had...

chapter 15: chapter 15 the goblin revenge early next morning before the other two were awake harry left the tent...

chapter 16: chapter 16 Godric's Hollow when harry woke the following day it was several seconds before he remembered...

chapter 17: chapter 17 Bathilda's secret harry stopped what was wrong they had only just reached the grave of the unknown...

chapter 18: chapter 18 the life and lies of albus Dumbledore the sun was coming up the pure colorless vastness of...

chapter 19: chapter 22 what caused this abrupt rupture had Dumbledore come to his senses had he told Grindelwald...

chapter 20: chapter 19 the silver doe it was snowing by the time Hermione took over the watch at midnight harry ...

chapter 21: chapter 20 Xenophilius Lovegood harry had not expected Hermione's anger to abate overnight and was the...

chapter 22: chapter 21 the tale of the three brothers harry turned to look at ron and Hermione neither of them said...

chapter 23: chapter 22 the Deathly Hallows harry fell panting onto grass and scrambled up at once they seemed to...

chapter 24: chapter 23 Malfoy Manor harry looked around at the other two now mere outlines in the darkness he said...

chapter 25: chapter 24 the wandmaker it was like sinking into an old nightmare for an instant harry knelt again ...

chapter 26: chapter 25 Shell Cottage Bill and Fleur's cottage stood alone on a cliff overlooking the sea its walls...

chapter 27: chapter 26 Gringotts their plans were made their preparations complete in the smallest bedroom a sin...

chapter 28: chapter 27 the final hiding place there was no means of steering the dragon could not see where it was...

chapter 29: chapter 28 the missing mirror harry's feet touched road he saw the achingly familiar Hog's Head high street...

chapter 30: chapter 29 the lost diadem Neville what the hell but Neville had spotted ron and Hermione and with ye...

chapter 31: chapter 30 the sacking of severus snape the moment her finger touched the mark harry scar burned sav...

chapter 32: chapter 31 the battle of hogwarts the enchanted ceiling of the great hall was dark and scattered wit...

chapter 33: chapter 32 the elder wand the world had ended so why had the battle not ceased the castle fallen sil...

chapter 34: chapter 33 the prince tale harry remained kneeling at snape side simply staring down at him until qu...

chapter 35: chapter 34 the forest again finally the truth lying with his face pressed into the dusty carpet of t...

chapter 36: chapter 35 king cross he lay facedown listening to the silence he was perfectly alone nobody was wat...

chapter 37: chapter 36 the flaw in the plan he was lying facedown on the ground again the smell of the forest fi...

แต่คิดวิธีที่ดีกว่าได้คือการแบ่ง chapter แบบไล่ลำดับ :)

comparable length each book and all book

In [296...

```
import re

# เก็บข้อมูล chapter ที่สั้นที่สุดและยาวที่สุดของทุกเล่ม
shortest_chapter_overall = {"book": None, "chapter": None, "words": float('inf')}
longest_chapter_overall = {"book": None, "chapter": None, "words": 0}

for i, book in enumerate(books, 0):
    # สร้าง pattern ไล่ลำดับ
    chapters = []
    current_pos = 0
    chapter_num = 1

    while True:
        next_chapter_pattern = f'chapter {chapter_num}'
        match = re.search(next_chapter_pattern, book[current_pos:], re.IGNORECASE)
        if not match:
            remaining_text = book[current_pos:].strip()
            if remaining_text:
                chapters.append(remaining_text)
            break
        start = current_pos + match.start()

        if chapter_num > 1:
            chapters.append(book[current_pos:start].strip())

        current_pos = start
        chapter_num += 1

    if i > 0:
        print(f"book_{i} has {len(chapters)} chapters")

    chapter_lengths = [len(ch.split()) for ch in chapters]

    for j, length in enumerate(chapter_lengths, 1):
        print(f"chapter {j} has {length} words")
        # update the shortest chapter
        if length < shortest_chapter_overall["words"]:
            shortest_chapter_overall = {"book": i, "chapter": j, "words": length}
        # update the longest chapter
```

```
        if length > longest_chapter_overall["words"]:
            longest_chapter_overall = {"book": i, "chapter": j, "words": len

# Check chapter is the shortest and which is the longest each books
min_length = min(chapter_lengths)
max_length = max(chapter_lengths)
min_index = chapter_lengths.index(min_length) + 1
max_index = chapter_lengths.index(max_length) + 1

if min_length != max_length:
    print(f" -> Shortest chapter in this book: chapter {min_index} ({mi
    print(f" -> Longest chapter in this book: chapter {max_index} ({max

print()

# Check chapter is the shortest and which is the longest all books
print(f"Overall shortest chapter: book_{shortest_chapter_overall['book']}, "
      f"chapter {shortest_chapter_overall['chapter']} "
      f"({shortest_chapter_overall['words']} words)")

print(f"Overall longest chapter: book_{longest_chapter_overall['book']}, "
      f"chapter {longest_chapter_overall['chapter']} "
      f"({longest_chapter_overall['words']} words)")
```

book_1 has 18 chapters

chapter 1 has 2560 words
chapter 2 has 2843 words
chapter 3 has 4487 words
chapter 4 has 5733 words
chapter 5 has 5395 words
chapter 6 has 4521 words
chapter 7 has 4529 words
chapter 8 has 4360 words
chapter 9 has 5085 words
chapter 10 has 5282 words
chapter 11 has 5937 words
chapter 12 has 5392 words
chapter 13 has 5441 words
chapter 14 has 3911 words
chapter 15 has 4634 words
chapter 16 has 5626 words
chapter 17 has 5341 words
chapter 18 has 4131 words

-> Shortest chapter in this book: chapter 1 (2560 words)

-> Longest chapter in this book: chapter 11 (5937 words)

book_2 has 36 chapters

chapter 1 has 3083 words
chapter 2 has 4038 words
chapter 3 has 3291 words
chapter 4 has 5349 words
chapter 5 has 5813 words
chapter 6 has 6535 words
chapter 7 has 6498 words
chapter 8 has 6134 words
chapter 9 has 4104 words
chapter 10 has 6594 words
chapter 11 has 5517 words
chapter 12 has 6053 words
chapter 13 has 5810 words
chapter 14 has 4182 words
chapter 15 has 6935 words
chapter 16 has 5184 words
chapter 17 has 5391 words
chapter 18 has 3445 words
chapter 19 has 6618 words
chapter 20 has 4423 words
chapter 21 has 4802 words
chapter 22 has 5797 words
chapter 23 has 7786 words
chapter 24 has 6651 words
chapter 25 has 4312 words
chapter 26 has 6809 words
chapter 27 has 2628 words
chapter 28 has 4473 words
chapter 29 has 4515 words
chapter 30 has 4844 words
chapter 31 has 7937 words
chapter 32 has 5485 words
chapter 33 has 8113 words
chapter 34 has 3864 words
chapter 35 has 4862 words
chapter 36 has 8982 words

-> Shortest chapter in this book: chapter 27 (2628 words)

-> Longest chapter in this book: chapter 36 (8982 words)

book_3 has 37 chapters

chapter 1 has 4121 words
chapter 2 has 2840 words
chapter 3 has 3179 words
chapter 4 has 2990 words
chapter 5 has 3774 words
chapter 6 has 2386 words
chapter 7 has 5259 words
chapter 8 has 5804 words
chapter 9 has 7175 words
chapter 10 has 3165 words
chapter 11 has 3235 words
chapter 12 has 5446 words
chapter 13 has 3896 words
chapter 14 has 4840 words
chapter 15 has 5247 words
chapter 16 has 6078 words
chapter 17 has 4066 words
chapter 18 has 6607 words
chapter 19 has 6342 words
chapter 20 has 7081 words
chapter 21 has 5556 words
chapter 22 has 4450 words
chapter 23 has 8142 words
chapter 24 has 6181 words
chapter 25 has 5501 words
chapter 26 has 8008 words
chapter 27 has 6989 words
chapter 28 has 7303 words
chapter 29 has 4334 words
chapter 30 has 6253 words
chapter 31 has 7910 words
chapter 32 has 2007 words
chapter 33 has 3931 words
chapter 34 has 2892 words
chapter 35 has 5819 words
chapter 36 has 6133 words
chapter 37 has 5100 words

-> Shortest chapter in this book: chapter 32 (2007 words)

-> Longest chapter in this book: chapter 23 (8142 words)

book_4 has 30 chapters

chapter 1 has 5038 words
chapter 2 has 4810 words
chapter 3 has 4716 words
chapter 4 has 5892 words
chapter 5 has 5883 words
chapter 6 has 5875 words
chapter 7 has 6464 words
chapter 8 has 4275 words
chapter 9 has 5841 words
chapter 10 has 5944 words
chapter 11 has 5095 words
chapter 12 has 5360 words
chapter 13 has 5351 words
chapter 14 has 6467 words
chapter 15 has 5763 words
chapter 16 has 6059 words

chapter 17 has 6311 words
chapter 18 has 6487 words
chapter 19 has 6174 words
chapter 20 has 6503 words
chapter 21 has 5703 words
chapter 22 has 5739 words
chapter 23 has 5701 words
chapter 24 has 5903 words
chapter 25 has 4970 words
chapter 26 has 6261 words
chapter 27 has 4448 words
chapter 28 has 3583 words
chapter 29 has 5434 words
chapter 30 has 5587 words
-> Shortest chapter in this book: chapter 28 (3583 words)
-> Longest chapter in this book: chapter 20 (6503 words)

book_5 has 38 chapters

chapter 1 has 5750 words
chapter 2 has 6019 words
chapter 3 has 5223 words
chapter 4 has 5670 words
chapter 5 has 5345 words
chapter 6 has 6748 words
chapter 7 has 4564 words
chapter 8 has 4172 words
chapter 9 has 7993 words
chapter 10 has 5780 words
chapter 11 has 5950 words
chapter 12 has 8294 words
chapter 13 has 8829 words
chapter 14 has 8039 words
chapter 15 has 6719 words
chapter 16 has 5877 words
chapter 17 has 6570 words
chapter 18 has 6854 words
chapter 19 has 6666 words
chapter 20 has 5987 words
chapter 21 has 7291 words
chapter 22 has 7978 words
chapter 23 has 7368 words
chapter 24 has 8007 words
chapter 25 has 8054 words
chapter 26 has 8499 words
chapter 27 has 7427 words
chapter 28 has 7973 words
chapter 29 has 7401 words
chapter 30 has 8219 words
chapter 31 has 7943 words
chapter 32 has 6126 words
chapter 33 has 3840 words
chapter 34 has 5220 words
chapter 35 has 7728 words
chapter 36 has 3786 words
chapter 37 has 7902 words
chapter 38 has 7759 words
-> Shortest chapter in this book: chapter 36 (3786 words)
-> Longest chapter in this book: chapter 13 (8829 words)

book_6 has 22 chapters

```
chapter 1 has 3683 words
chapter 2 has 3840 words
chapter 3 has 4357 words
chapter 4 has 5077 words
chapter 5 has 6533 words
chapter 6 has 6672 words
chapter 7 has 4213 words
chapter 8 has 5008 words
chapter 9 has 5243 words
chapter 10 has 7009 words
chapter 11 has 5529 words
chapter 12 has 4750 words
chapter 13 has 4226 words
chapter 14 has 5497 words
chapter 15 has 5484 words
chapter 16 has 4343 words
chapter 17 has 4164 words
chapter 18 has 2237 words
chapter 19 has 5056 words
chapter 20 has 1924 words
chapter 21 has 7210 words
chapter 22 has 4892 words
-> Shortest chapter in this book: chapter 20 (1924 words)
-> Longest chapter in this book: chapter 21 (7210 words)
```

```
book_7 has 17 chapters
chapter 1 has 4569 words
chapter 2 has 3436 words
chapter 3 has 3833 words
chapter 4 has 3640 words
chapter 5 has 6511 words
chapter 6 has 6235 words
chapter 7 has 4443 words
chapter 8 has 3042 words
chapter 9 has 4896 words
chapter 10 has 4275 words
chapter 11 has 3311 words
chapter 12 has 5454 words
chapter 13 has 3171 words
chapter 14 has 3444 words
chapter 15 has 5096 words
chapter 16 has 6371 words
chapter 17 has 5355 words
-> Shortest chapter in this book: chapter 8 (3042 words)
-> Longest chapter in this book: chapter 5 (6511 words)
```

```
Overall shortest chapter: book_6, chapter 20 (1924 words)
Overall longest chapter: book_2, chapter 36 (8982 words)
```

Find the characters

In [297...

```
characters = ['harry', 'ron', 'hermione', 'dobby', 'malfoy', 'voldemort|you know
for name in characters:
    matches = re.findall(name, cleaned_text)
    print(f"{name.upper()} occurences is {len(matches)} times.")
```

HARRY occurrences is 18254 times.
RON occurrences is 8200 times.
HERMIONE occurrences is 5365 times.
DOBBY occurrences is 470 times.
MALFOY occurrences is 1363 times.
VOLDEMORT|YOU KNOW WHO occurrences is 1266 times.
SNAPE occurrences is 1829 times.
DUMBLEDORE occurrences is 3375 times.
HAGRID occurrences is 2044 times.
SIRIUS occurrences is 1137 times.

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